

Invariant subgroup: Subgroup N s.t. for $t \in N$, $\forall r \in G$ s.t. $rti^{-1} \in N$

→ Each component of a direct product group is an invariant subgroup.

ex: $SU(2) \times U(1) = G$, $a \in SU(2)$, $b \in U(1)$

$$(a, b_1) \cdot_G (a_2, b_2) := (a_1 a_2) (b_1 b_2)$$

Let $H_1 = SU(2)$, $h \in H_1$ and $a \in G$, $a = a_1 b_1$

note $h = h \cdot 1_{U(1)}$

$$\begin{aligned} a(h) (a_1^{-1} b_1^{-1}) &= a(h a_1^{-1} b_1^{-1}) = (a_1 h) (h a_1^{-1} b_1^{-1}) \\ &= (a_1 h a_1^{-1}) \cdot 1_{U(1)} = a_1 h a_1^{-1} \in SU(2) \quad \checkmark \end{aligned}$$

Same for $U(1)$.

→ Direct prod. underlying set is a cart. product

$$(G, *) , (M, \Delta) : G \times M$$

$$(g_1, m_1) \cdot (g_2, m_2) := (g_1 * g_2, m_1 \Delta m_2)$$

Simple group: Cannot be decomposed into invariant subgroups i.e.

has no non-trivial invariant subgroups

• ex $U(n)$ is not simple $U(n) = SU(n) \times U(1)$ direct

Semi-simple groups: Groups which decompose into the product of simple groups

→ $U(n)$ is semi simple

ex of Lie group: $a(\vec{\theta}) = a(\theta_1, \theta_2, \dots, \theta_n)$ and $e = a(\vec{0})$

$$a(\vec{\theta}) \cdot a(\vec{\phi}) = a(\vec{\xi})$$

corresponds to the mapping of the parameters onto itself?

$$f(\vec{\theta}, \vec{\phi}) = \vec{\xi}$$

$$f: V \times V \rightarrow V$$

$$f(0, \vec{\theta}) = f(\vec{\theta}, 0) = \vec{\theta} \mid f(\theta, f(\phi, \xi)) = f(f(\theta, \phi), \xi)$$

$$f(\theta, \theta') = 0 \text{ if } a(\theta)^{-1} \text{ is parameterized by } a(\theta')$$

$$\bullet a(\vec{\theta}) = \exp\{i \vec{\theta} \cdot \vec{X}\} = a(0) + i \theta_k X_k + \dots$$

$$X_k = -i \frac{\partial a}{\partial \theta_k} \bigg|_{\theta=0} \text{ (group generators)} \quad (\text{near } 1)$$

* The commutator of group elements is (for $a(\phi) \neq a(\theta)$)

$$a(\phi) a(\theta) a(\phi)^{-1} a(\theta)^{-1}$$

For this to make sense the above must close in g , i.e.

$$= a(\vec{\xi}) \text{ with } \vec{\xi} = \theta \cdot (A \phi) \quad a(\theta \phi) = a(\theta, \phi) = 0$$

Let $\phi = 0$, then

$$\begin{cases} e^{a(\phi)} e^{a(\phi)^{-1}} = e = a(\xi) \text{ near } \xi = 0 \\ \text{or } \xi = g(0, \phi) = 0 \text{ and likewise } g(\phi, 0) = 0 \text{ as claimed.} \end{cases}$$

expand ξ :

$$\partial: g_i(\phi, \phi) = g_i(0, \phi) + \theta_k \frac{\partial g_i(\phi, \phi)}{\partial \theta_k} \bigg|_{\phi=0} + \theta_k \theta_l \frac{\partial^2 g_i}{\partial \theta_k \partial \theta_l} + \dots$$

$$\text{both: } g_i(\omega, \phi) = \theta_k \theta_l \left(\frac{\partial}{\partial \theta_k} \frac{\partial}{\partial \theta_l} g_i \right) + \dots$$

note all other terms vanish from the boundary conditions
 $\xi^i = C_{ke}^i \theta_k \phi_e + \dots$

$$a(\xi) = e + i \xi_i X_i + \dots$$

$$a(\xi) = e + i C_{ke}^i \theta_k \phi_e X_i + \dots$$

$$a(\phi) a(\theta) a(\phi)^{-1} a(\theta)^{-1} = e + \theta_j \theta_k [X_j, X_k] + \dots$$

$$\rightarrow \text{Lie Alg. } [X_j, X_k] = i C_{jk}^l X_l$$

$$\text{Diacol} = e^{i\theta \cdot T}, \quad T(X_j) \equiv T_j$$

U(1): Electro

SU(2): isospin (strong)

SU(3): "eight fold way" color force (weak)

$$\text{Unit: } \{M \mid M^\dagger M = M M^\dagger = I, \det(M) = 1\}$$

$$\bullet \text{ Note } M = e^{iH} \Rightarrow (I - iH^\dagger)(I + iH) = I$$

$$I - iH + iH^\dagger + \mathcal{O}(H^2) = I \Rightarrow H = H^\dagger$$

H is Hermitian

$$\det(e^H) = e^{\text{tr}(H)} = 1 \Rightarrow \text{tr}(H) = 0$$

H: Hermitian & traceless

• Thus H has $n^2 - 1$ elements, each of which are real valued

→ Expand H via spectral thm over a basis:

$$H = \epsilon_a J^a$$

ϵ_a : real group parameters (possibly eigenvalues)

J^a : Group basis (generators)

Note that

$$n^2 - 1 = (n-1)(n+1)$$

$$\text{Tr}(H) = \epsilon_a \text{Tr}(J^a)$$

Since trace is linear:

$$\text{Tr}(AB) = \text{Tr}(A+B) = (A+B)_{ii} = A_{ii} + B_{ii}$$

$$\text{and so } \text{Tr}(J^a) = 0 \quad \text{and } \text{Tr}(H) \text{Tr}(B)$$

Hermitian: $H^\dagger = H$

Rank: Dimension of the Cartan subalgebra which SU(2): 3 group parameters: $2^2 - 1$

$$U(\epsilon_1, \epsilon_2, \epsilon_3) = e^{i\epsilon_a \sigma_a}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$[J_a, J_b] = i f_{ab}^c J_c, \quad J_i = \frac{\sigma_i}{2}$$

$$[J_a, J_b] = i \epsilon_{abc} J_c$$

* Reps of SU(2): SU(2) & SO(3) have the same alg.

$$\text{SO}(3)/\mathbb{Z} \cong \text{SU}(2)$$

$$\vec{J}^2 = J_1^2 + J_2^2 + J_3^2 \quad (\text{Casimir operator})$$

which commutes with every generator

$$[J^2, J_a] = 0 \quad a = 1, 2, 3$$

$$\sigma_1^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \sigma_3^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\sigma_2^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{so } \hat{J}^2 = \frac{3}{4} \hat{1}$$

$$J_3 = J_z$$

• Ladder operators: $J_{\pm} := J_1 \pm i J_2$

$$J^2 = \frac{1}{2} \{J_+, J_-\} + J_z^2$$

$$\begin{aligned} [J_+, J_-] &= [(J_1 + i J_2), (J_1 - i J_2)] \\ &= -i [J_1, J_2] + i [J_2, J_1] = -2i [J_1, J_2] = -2i (i \epsilon_{123} J_3) \end{aligned}$$

$$\therefore [J_+, J_-] = 2 J_z$$

$$[J_{\pm}, J_z] = \mp J_{\pm}$$

$$* J^2 |\lambda, m\rangle = \lambda |\lambda, m\rangle$$

$$J_z |\lambda, m\rangle = m |\lambda, m\rangle$$

From above $J_{\pm} |\lambda, m\rangle$ are also eigen states of J_z with $m = \pm 1$

$$\text{so } J_{\pm} J_z |\lambda, m\rangle = J_{\pm} m |\lambda, m\rangle = m J_{\pm} |\lambda, m\rangle$$

$$= (J_z J_{\pm} \mp J_{\pm}) |\lambda, m\rangle = (J_z J_{\pm} |\lambda, m\rangle \mp J_{\pm} |\lambda, m\rangle)$$

$$J_z (J_{\pm} |\lambda, m\rangle) = (m \pm 1) (J_{\pm} |\lambda, m\rangle)$$

$$\text{Hence } J_{\pm} |\lambda, m\rangle = C_{\pm}(\lambda, m) |\lambda, m \pm 1\rangle$$

for a given total magnetic quantum number λ , the possible values of m are bounded via

$$J^2 - J_z^2 = J_1^2 + J_2^2 \geq 0$$

$$\lambda - m^2 \geq 0$$

- Let j be the largest value of m i.e. the highest weight state is given by

$$J_+ |\lambda, j\rangle = 0$$

$$\begin{aligned} 0 &= J_- J_+ |\lambda, j\rangle = (2J^2 - 2J_z^2 - J_+ J_-) |\lambda, j\rangle \\ &= 2(J^2 - J_z^2 - J_z) |\lambda, j\rangle \\ &= 2(\lambda - j^2 - j) |\lambda, j\rangle = 0 \end{aligned}$$

$$\lambda - j^2 - j = 0$$

or $\lambda = j(j+1)$ j : largest value of J_z : 0

- let j' be the smallest value of m s.t. $J_- |\lambda, j'\rangle \neq 0$

$$\text{then } \lambda = j'(j'-1)$$

$$\Rightarrow j(j+1) = j'(j'-1)$$

Solutions: $j = -j'$ $\wedge$ $j' = j+1$ $\leftarrow$ Contradiction

$$\text{so } j' = -j$$

- $j - j' = 2j$ must be an integer

so j can be of the form $j = \frac{n}{2}$

$$J_+ |\lambda, m\rangle = C_+ |\lambda, m+1\rangle$$

$$\langle \lambda, m | J_- J_+ |\lambda, m\rangle = |C_+ (\lambda, m)|^2$$

$$= \langle \lambda, m | (J^2 - J_z^2 - J_z) |\lambda, m\rangle = (j(j+1) - m^2 - m)$$

$$C_+ (\lambda, m) = ((j-m)(j+m+1))^{1/2}$$

$$C_- (\lambda, m) = ((j+m)(j-m+1))^{1/2}, \quad m = j, \dots, -j$$

dim of the rep is all the values of m :

$$2j+1$$

ex: $J = \frac{1}{2}$, $m = \pm \frac{1}{2}$, $J_z |\frac{1}{2}, \pm \frac{1}{2}\rangle = \pm \frac{1}{2} |\frac{1}{2}, \pm \frac{1}{2}\rangle$

$$|\frac{1}{2}, \frac{1}{2}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|\frac{1}{2}, -\frac{1}{2}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$J_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$$

$$c=0, \quad a=\frac{1}{2}$$

$$J_z \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & b \\ 0 & d \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} \quad d = -\frac{1}{2}$$

$$J_z = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{2} \sigma_z$$

$$J_+ |1/2, 1/2\rangle = 0 \rightarrow J_+ \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad J_+ = \begin{pmatrix} 0 & b \\ 0 & c \end{pmatrix}$$

$$J_+ |1/2, -1/2\rangle = |1/2, 1/2\rangle \Rightarrow \begin{pmatrix} 0 & b \\ 0 & c \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow c=0, b=1$$

$$J_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad J_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad J_x = J_+$$

$$J_1 = \frac{1}{2} \sigma_x, \quad J_2 = \frac{1}{2} \sigma_y$$

ex 2: $|j\rangle = 1, \quad m = 1, 0, -1, \quad \lambda = |j|$

only z direction - which we can always boost to

$$|1, 1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1, 0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1, -1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$J_z |1, m\rangle = m |1, m\rangle \quad J_z = \begin{pmatrix} a & b & c \\ b & e & g \\ c & g & f \end{pmatrix}$$

$$\begin{pmatrix} & & \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad a=1, \quad b=0, \quad c=0$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & e & g \\ 0 & g & f \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad g=0, \quad e=0$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & f \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad f=-1$$

$$J_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \text{can find all others similarly}$$

SU(2) product reps: (tensor reps)

Two electrons: $J=1$ or $J=0$

Tensor Method of SU(n):

$$| \mathbb{C}^n \supset \psi_i$$

$$SU(n) \oplus \overline{SU(n)}$$

$$\psi_i = (\psi_1, \dots, \psi_n) \mid \psi'_i = U_{ij} \psi_j, \quad U \in SU(n)$$

for ψ^* , basis

so $(\psi, \phi) := \psi_i^* \phi_i$ which is SU(n) invariant

U invariant.

$$\psi_i^* \mapsto (\psi'_i)^* = (U_{ij} \psi_j)^* = \psi_j^* U_{ij}^* = \psi_j^* U_{ji}^\dagger$$

• Upper/lower indices:

$$\psi^i \equiv \psi_i^*, \quad U^i_j = U_{ji}^*$$

$$U_i^i = U_{ii}$$

$$\psi_i \mapsto \psi'_i = U_{ij} \psi_j \quad \text{for } SU(n) \rightarrow$$

$$\psi^i \mapsto U^{ij} \psi_j \quad \text{for } \overline{SU(n)} \swarrow$$

$$\begin{aligned} \langle \psi, \phi \rangle &= \langle U\psi, U\phi \rangle = (U\psi)_i^* (U\phi)_i = \psi_j^* U_{ij}^* U_{ik} \psi_k = \psi_j^* U_{ij}^* U_{ik} \psi_k = \psi_j^* \delta_{jk} \psi_k = \psi_j^* \psi_j = \psi^i \psi_i \end{aligned}$$

$$\therefore U^i_j U_i^k = \delta_j^k$$

* ψ_i : Basis for the defining rep of $SU(n)$ $\frac{n}{1}$

$\bar{\psi}_i$: Basis for the conjugate rep of $SU(n)$ $\frac{n}{1}$

↳ Note $\mathbb{I}$ or δ_j^i is $SU(n)$ invariant:

$$\delta_j^i \mapsto \bar{p}_n^i \delta_\ell^k p_j^\ell = \bar{p}_\ell^i p_j^\ell = \delta_j^i \quad \checkmark$$

↳ Levi-Civita symbol is the $SU(n)$ "metric" and is invariant

Note: $\det(A) \epsilon_{a_1 \dots a_n} = \epsilon_{b_1 \dots b_n} A_{a_1}^{b_1} \dots A_{a_n}^{b_n}$

proof: $\det(A) = A_1^{a_1} \dots A_n^{a_n} \epsilon_{a_1 \dots a_n}$

$$= A_1^{[a_1} \dots A_n^{a_n]} \epsilon_{a_1 \dots a_n}$$

$$= A_{[1}^{a_1} \dots A_n^{a_n]} \epsilon_{a_1 \dots a_n}$$

$$= \delta_{[1}^{b_1} \dots \delta_n^{b_n]} A_{b_1}^{a_1} \dots A_{b_n}^{a_n} \epsilon_{a_1 \dots a_n}$$

$$= \delta_{1 \dots n}^{b_1 \dots b_n} A_{b_1}^{a_1} \dots A_{b_n}^{a_n} \epsilon_{a_1 \dots a_n}$$

$$= \frac{1}{n!} \epsilon^{b_1 \dots b_n} A_{b_1}^{a_1} \dots A_{b_n}^{a_n} \epsilon_{a_1 \dots a_n}$$

Start from

$$\det(A) \epsilon_{a_1 \dots a_n} = \epsilon_{b_1 \dots b_n} A_{a_1}^{b_1} \dots A_{a_n}^{b_n}$$

mult. by $\epsilon^{a_1 \dots a_n}$

$$\det(A) n! = \epsilon_{b_1 \dots b_n} \epsilon^{a_1 \dots a_n} A_{a_1}^{b_1} \dots A_{a_n}^{b_n}$$

$$\det(A) = \frac{1}{n!} \epsilon \dots \quad \text{as above.}$$

Hence

$$\epsilon_{i_1 i_2 \dots i_n} = U_{i_1}^{j_1} \dots U_{i_n}^{j_n} \epsilon_{j_1 j_2 \dots j_n} = \det(U) \epsilon_{i_1 \dots i_n} = \epsilon_{i_1 \dots i_n}$$

so ϵ is invariant.

* $\epsilon_{i_1 \dots i_n} \psi_{i_2 \dots i_n} = \psi_{i_1}$

$SU(n)$ invariant scalar $\psi \equiv \epsilon_{i_1 \dots i_n} \psi^{i_1 \dots i_n}$

Irreducible reps and the Young tableaux:

P_{12} : permutation operator

$$P_{12}(\psi_{ij}) = \psi_{ji}$$

$$P_{12}(\psi^{ij}) = U_k^i U_l^j P_{12} \psi^{kl}$$

Form eigenstates of P_{12}

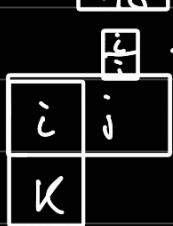
$$\psi_{[ij]} = \frac{1}{2}(\psi_{ij} - \psi_{ji})$$

$$\psi_{[ij]} = \frac{1}{2}(\psi_{ij} - \psi_{ji})$$

$$P_{12} \psi_{[ij]} = -\psi_{[ij]}$$

$$P_{12} \psi_{[ij]} = -\psi_{[ij]}$$

$$[ij] = \psi_{[ij]}$$



$$\psi_{[ij]}$$

$$\psi_{ijk} = \frac{1}{2}(\psi_{ijk} + \psi_{jik})$$

$$= \frac{1}{4}(\psi_{ijk} - \psi_{kji} + \psi_{jik} - \psi_{jki})$$

$$= \frac{1}{4}(\psi_{ijk} + \psi_{jik} - \psi_{kji} - \psi_{kij} + \psi_{jik} + \psi_{ijk} \dots)$$

$$\equiv \psi_{ijk} + \psi_{jik} - \psi_{kji} - \psi_{kij}$$

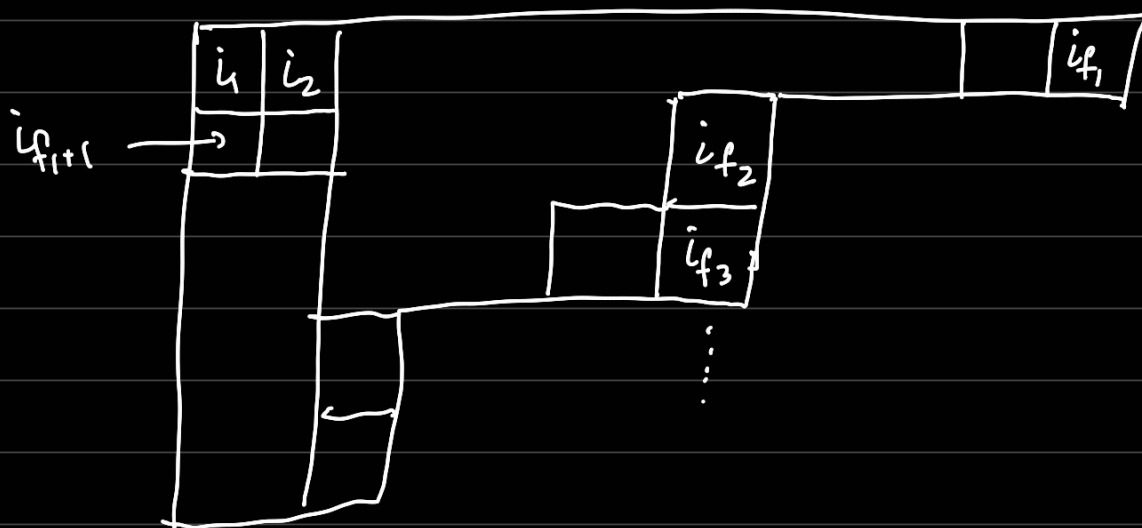
$$f_1 \geq f_2 \geq \dots \quad f_1 + f_2 + \dots = f$$

f : total Boxes

each box has an index $i_k = 1, 2, \dots, n$

each tableau: tensor

$$\psi_{i_1, i_2, \dots, i_{f_1}, i_{f_1+1}, \dots, i_{f_1+f_2}}$$



(i) Indices appearing in the same row are first subject to symmetrization

(ii) Subsequent indices appearing in the same column are subject to antisymmetrization

* Standard tableau: Increases $L \rightarrow R$

ex: $n=3$



Try mixed-symmetry tensor



Standard tableaux

1 1
2

1 1
3

1 2
2

1 3 1 3
2 3

2 2 2 3
3 3

Non-standard

1 1
1

2 3
1

2 1
3

yield tensors which vanish or are not independent of standard tab.

* Thus # of independent tensors is equal to the number of standard tableaux which can be formed.

$$i_k = 1, \dots, n$$

ex:
(a) $\left. \begin{array}{|c|} \hline \vdots \\ \hline \end{array} \right\} k \quad \binom{n}{k}$

• number of ways of choosing k distinct elements out of n

(b) $\underbrace{\boxed{1 \dots 1}}_k : \binom{n}{k}$